

01. $x^2 - 16x + 64$.

$$\begin{array}{r} 1 \times 64 \\ 2 \times 32 \\ \hline 8 \times 8 \end{array}$$

$$x^2 - 8x - 8x + 64$$

$$\begin{array}{cc} x(x - 8) & - 8(x - 8) \\ (x - 8) & (x - 8) \end{array}$$

02. $x^2 + 22x + 121$.

$$\begin{array}{r} 1 \times 121 \\ 11 \times 11 \\ \hline 11 \times 11 \end{array}$$

$$x^2 + 11x + 11x + 121$$

$$\begin{array}{cc} x(x + 11) & + 11(x + 11) \\ (x + 11) & (x + 11) \end{array}$$

$$03. x^2 - x - 72.$$

$$\begin{array}{l} 1 \times 72 \\ 2 \times 36 \\ 9 \times 8 \end{array}$$

$$\cancel{x} \quad x^2 - 9x + 8x - 72.$$

$$x(x - 9) + 8(x - 9)$$

$$(x + 8)(x - 9)$$

$$04. x^2 - 3x - 18$$

$$\begin{array}{l} 1 \times 18 \\ 2 \times 14 \\ \boxed{6 \times 3} \end{array}$$

$$x^2 - 6x + 3x - 18$$

$$x(x - 6) + 3(x - 6)$$

$$(x + 3)(x - 6)$$

$$05. x^2 - 4x - 45$$

$$\begin{array}{l} 1 \times 45 \\ \boxed{9 \times 5} \end{array}$$

$$x^2 - 9x + 5x - 45$$

$$x(x - 9) + 5(x - 9)$$

$$(x - 9)(x + 5)$$

$$06 \quad x^2 - 16x + 63.$$

$$\begin{array}{r} 1 \times 63 \\ \hline 7 \times 9 \end{array}$$

$$x^2 - 7x - 9x + 63$$

$$x(x - 7) - 9(x - 7)$$

$$(x - 9)(x - 7)$$